as an IoT protocol. It is used to exchange data between clients and the server. It is a lightweight messaging protocol.

This protocol has many advantages:

- It is small in size and has low power usage.
- It is a lightweight protocol.
- It is based on low network usage.
- It works entirely in real-time.

Considering all the above reasons, MQTT emerges as the perfect IoT data protocol.

How MQTT works: MQTT is based on a client-server relationship. The server manages the requests that come from different clients and sends the required information to clients. MQTT is based on two operations.

i) *Publish:* When the client sends data to the MQTT broker, this operation is known as 'Publish'.

ii) *Subscribe:* When the client receives data from the broker, this operation is known as'Subscribe'.

The MQTT broker is the mediator that handles these operations, primarily taking messages and delivering them to the application or client.

Let's look at the example of a device temperature sensor, which sends readings to the MQTT broker, and then information is delivered to desktop or mobile applications. As stated earlier, 'Publish' means sending readings to the MQTT broker and 'Subscribe' means delivering the information to the desktop/mobile application.

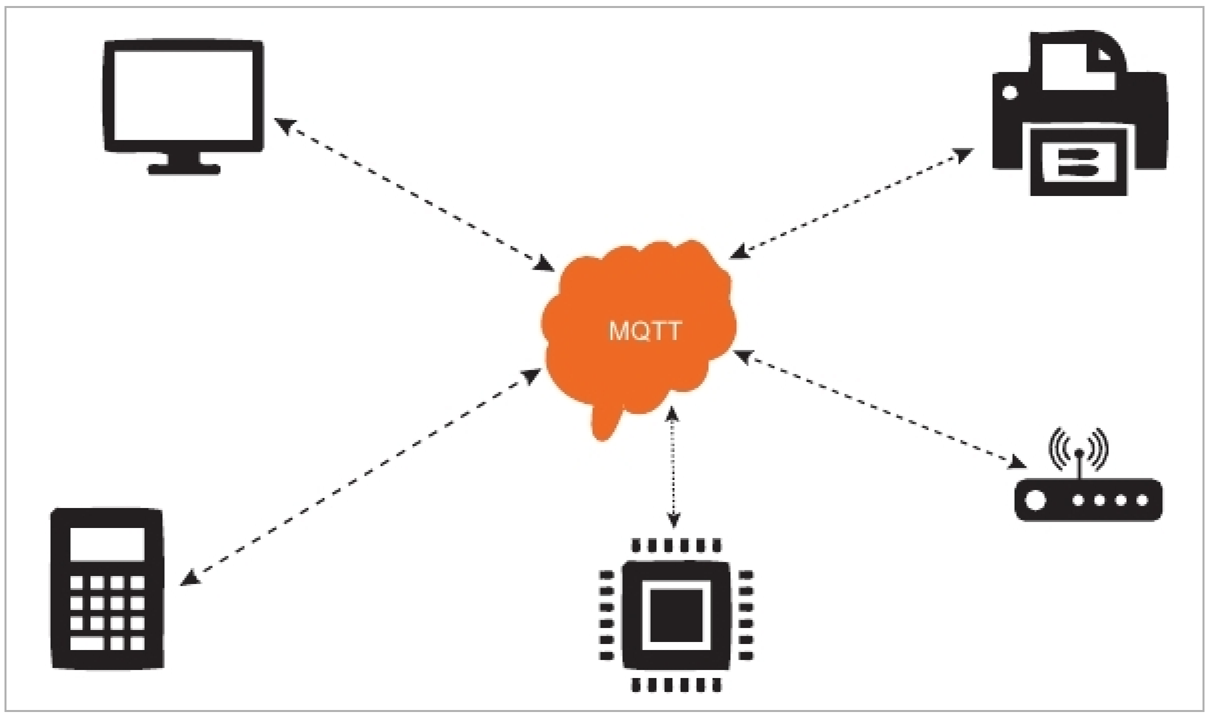

Figure 2: Messaging Queuing Telemetry Transmit protocol

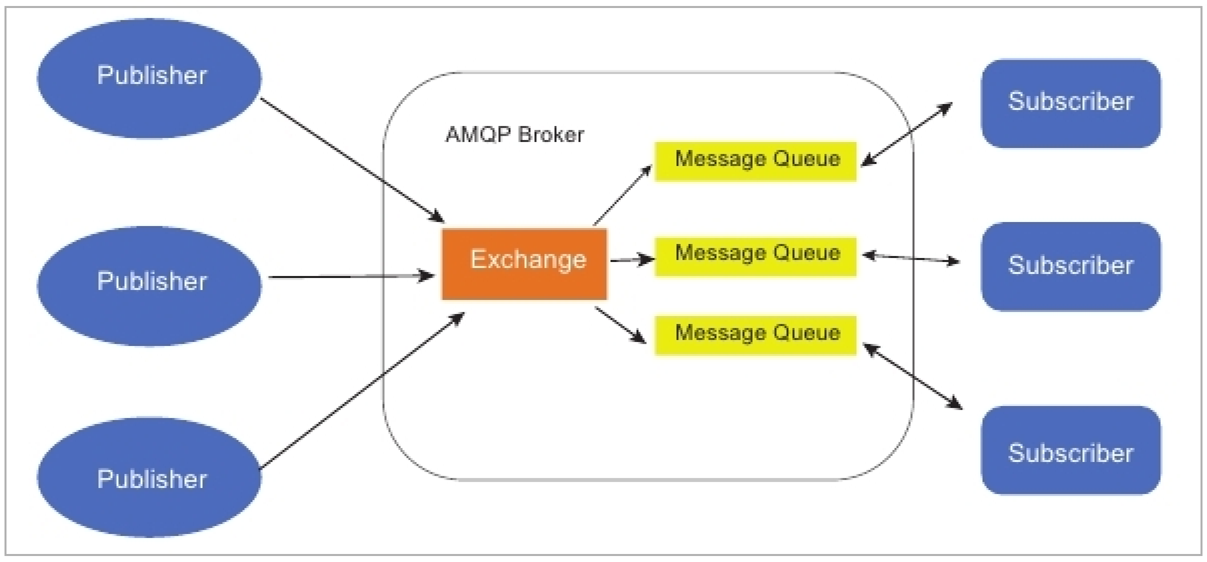

Figure 3: Advance Message Queuing Protocol

AMQP

Advanced Message Queuing Protocol is a peer-to-peer protocol, where one peer plays the role of the client application and the other peer plays the role of the delivery service or broker. It is the combination of hard and fast components that basically routes and saves messages within the delivery service or broker carrier.

The benefits of AMQP are:

- It helps to send messages without them getting missed out.
- It helps to guarantee a 'one-time only' and secured delivery.
- It provides a secure connection.
- It always supports acknowledgments for message delivery or failure.

How AMQP works and its architecture: The AMQP architecture is made up of the following parts.

Exchange – Messages that come from the publisher are accepted by Exchange, which routes them to the message queue.

Message queue – This is the combination of multiple queues and is helpful for processing the messages.

Binding – This helps to maintain the connectivity between Exchange and the message queue.

The combination of Exchange and the message queues is known as the broker or AMQP broker.

Constrained Application Protocol (CoAP)

This was initially used as a machine-to-machine (M2M) protocol, and later began to be used as an IoT protocol. It is a Web transfer protocol that is used with constrained nodes and constrained networks. CoAP uses the RESTful architecture, just like the HTTP protocol.

The advantages CoAP offers are:

- It works as a REST model for small devices.
- As this is like HTTP, it's easy for developers to work on.
- It is a one-to-one protocol for transferring information between the client and server, directly.
- It is very simple to parse.

How CoAP works and its architecture: From Figure 4, we can understand that CoAP is the